

IP Addressing Exercises

Lecture 6, Spring 2026

IP Addressing

- Hierarchical Addressing
- Assigning Addresses
- Writing Addresses
- Aggregating Routes
- IPv6 Changes

Q. IP Addressing

Suppose hofstra.edu is the Provider AS for Engineering (Engr), CS, Math, and Chemistry, and needs to assign IPv4 addresses to them. Assume that the CIDR (Classless InterDomain Routing) addressing scheme is used, and that the domain hofstra.edu has the 123.100.0.0/16 prefix reserved. For each problem, choose the smallest unused prefix to assign next.

- 1) Which addresses are included in the Math department's prefix? How many addresses are in this range?
- 2) 123.100.192.0/18 is reserved for Engr and CS. Assign equal halves of this address space to the two departments.
- 3) You want to start a new department of AI, and assign it an unused address range. You foresee that no more than 50 people will enroll. Assuming one address per person, what prefix would you assign to it to minimize unused/wasted addresses?

